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VECTORS - ALL FORMULAS

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1. Vector

$$A = (x, y)$$

$$A = xi + yj + zk$$

Example:

$$A = 3i + 4j$$

2. Magnitude (Length)

$$|A| = \sqrt{x^2 + y^2}$$

3D:

$$|A| = \sqrt{x^2 + y^2 + z^2}$$

Example:

$$A = (3,4)$$

$$|A| = \sqrt{3^2 + 4^2}$$

$$= \sqrt{25}$$

$$= 5$$

3. Unit Vector

$$\hat{A} = A / |A|$$

Example:

$$A = (3,4)$$

$$|A| = 5$$

$$\hat{A} = (3/5, 4/5)$$

4. Position Vector

If $P(x,y)$

$$OP = xi + yj$$

Example:

$$P(5,2)$$

$$OP = 5i + 2j$$

5. Vector Addition

$$(x_1, y_1) + (x_2, y_2)$$

$$= (x_1+x_2, y_1+y_2)$$

Example:

$$(2,3) + (5,4)$$

$$= (7,7)$$

6. Vector Subtraction

$$(x_1, y_1) - (x_2, y_2)$$

$$= (x_1-x_2, y_1-y_2)$$

Example:

$$(7,5) - (2,1)$$

$$= (5,4)$$

7. Scalar Multiplication

$$k(x,y)$$

$$= (kx, ky)$$

Example:

$$3(2,4)$$

$$= (6,12)$$

8. Distance Between Two Points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example:

$$A(2,3)$$

$$B(6,6)$$

$$d = \sqrt{4^2 + 3^2}$$

$$= 5$$

9. Vector Between Two Points

$$\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1)$$

Example:

$$A(2,3)$$

$$B(5,7)$$

$$\overrightarrow{AB} = (3,4)$$

10. Dot Product

$$A \cdot B$$

$$= x_1x_2 + y_1y_2 + z_1z_2$$

Also,

$$A \cdot B$$

$$= |A||B|\cos\theta$$

Example:

$$A=(2,3)$$

$$B=(4,5)$$

$$A \cdot B$$

$$= 2 \times 4 + 3 \times 5$$

$$= 23$$

11. Angle Between Two Vectors

$$\cos\theta$$

$$= (A \cdot B) / (|A||B|)$$

Example:

$$A=(1,0)$$

$$B=(0,1)$$

$$\text{Dot Product} = 0$$

$$\theta = 90^\circ$$

12. Perpendicular Vectors

Condition:

$$A \cdot B = 0$$

Example:

$$(2,3)$$

$$(-3,2)$$

$$2(-3)+3(2)=0$$

13. Cross Product (3D)

$$\mathbf{A} \times \mathbf{B}$$

Magnitude:

$$|\mathbf{A} \times \mathbf{B}| \\ = |\mathbf{A}||\mathbf{B}|\sin\theta$$

Example:

$$\mathbf{A} = (1, 0, 0)$$

$$\mathbf{B} = (0, 1, 0)$$

$$\mathbf{A} \times \mathbf{B} = (0, 0, 1)$$

14. Area of Parallelogram

Area

$$= |\mathbf{A} \times \mathbf{B}|$$

15. Area of Triangle

Area

$$= \frac{1}{2} |\mathbf{A} \times \mathbf{B}|$$

16. Projection

Scalar Projection

$$(A \cdot B)/|B|$$

Vector Projection

$$\text{Proj}_B(A)$$

$$= ((A \cdot B)/|B|^2)B$$

17. Direction Cosines

$$l = x/|A|$$

$$m = y/|A|$$

$$n = z/|A|$$

Property

$$l^2 + m^2 + n^2 = 1$$

18. Section Formula

$$P\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$$

Example

$$A(2,3)$$

$$B(8,9)$$

Ratio 1:2

$$P=(4,5)$$

19. Midpoint Formula

$$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

Example

$$A(2,4)$$

B(6,8)

Midpoint=(4,6)

20. Collinear Condition

Area = 0

21. Vector Equation of Line

$r = a + tb$

Example

$r = (1,2) + t(3,4)$

22. Components

$A_x = r \cos\theta$

$A_y = r \sin\theta$

Example

$$r=10$$

$$\theta=30^\circ$$

$$A_x=10\cos 30^\circ$$

$$A_y=10\sin 30^\circ$$

23. Triangle Law

$$R=A+B$$

Magnitude

$$R=\sqrt{(A^2+B^2+2AB\cos\theta)}$$

Example

$$A=3$$

$$B=4$$

$$\theta=90^\circ$$

$$R=5$$

24. Parallelogram Law

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

Direction

$$\tan\alpha = (B\sin\theta)/(A + B\cos\theta)$$

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MOST IMPORTANT FORMULAS

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$$|A| = \sqrt{x^2 + y^2 + z^2}$$

$$\hat{A} = A/|A|$$

$$A \cdot B = x_1x_2 + y_1y_2 + z_1z_2$$

$$A \times B$$

$$\cos\theta = (A \cdot B)/(|A||B|)$$

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Midpoint} = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

$$\text{Area of Triangle} = \frac{1}{2} |\mathbf{A} \times \mathbf{B}|$$

$$\text{Area of Parallelogram} = |\mathbf{A} \times \mathbf{B}|$$

$$\text{Projection} = ((\mathbf{A} \cdot \mathbf{B})/|\mathbf{B}|^2)\mathbf{B}$$